

Adapting the European Union AI Act to deal with generative artificial intelligence

The European Union's draft AI Act already needs to be revised to account for the opportunities and harms of generative AI.

J. Scott Marcus

The author gratefully acknowledges extensive helpful feedback from Bertin Martens and Kai Zenner.

When the European Commission in April 2021 proposed an AI Act to establish harmonised EU-wide harmonised rules for artificial intelligence, the draft law might have seemed appropriate for the state of the art. But it did not anticipate OpenAI's release of the ChatGPT chatbot, which has demonstrated that AI can generate text at a level similar to what humans can achieve. ChatGPT is perhaps the best-known example of generative AI, which can be used to create texts, images, videos and other content.

Generative AI might hold enormous promise, but its risks have also been flagged up⁵. These include (1) sophisticated disinformation (eg deep fakes or fake news) that could manipulate public opinion, (2) intentional exploitation of minorities and vulnerable groups, (3) historical and other biases in the data used to train generative AI models that replicate stereotypes and could lead to output such as hate speech, (4) encouraging the user to perform harmful or self-harming activities, (5) job losses in certain sectors where AI could replace humans, (6) 'hallucinations' or false replies, which generative AI can articulate very convincingly, (7) huge computing demands and high energy use, (8) misuse by organised crime or terrorist groups, and finally, (9) the use of copyrighted content as training data without payment of royalties.

To address those potential harms, it will be necessary to come to terms with the foundation models that underlie generative AI. Foundation models, or models through which machines learn from data, are typically trained on vast quantities of unlabelled data, from which they infer patterns without human supervision. This unsupervised learning enables foundation models to exhibit capabilities beyond those originally envisioned by their developers (often referred to as 'emergent capabilities').

The evolving AI Act

The proposed AI Act (European Commission, 2021), which at time of writing is still to be finalised between the EU institutions⁶, is a poor fit for foundation models. It is structured around the idea that each AI application can be allocated to a risk category based on its intended use. This structure largely reflects traditional EU product liability legislation, in which a product has a single, well-defined purpose. Foundation models however can easily be customised to a great many potential uses, each of which has its own risk characteristics.

In the ongoing legislative work to amend the text, the European Parliament has proposed that providers of foundation models perform basic due diligence on their offerings. In particular, this should include:

- **Risk identification.** Even though it is not possible to identify in advance all potential use cases of a foundation model, providers are typically aware of certain vectors of risk. OpenAI knew, for instance, that the training dataset for GPT-4 featured certain language biases because over 60 percent of all websites are in English. The European Parliament would make it mandatory to identify and mitigate reasonably foreseeable risks, in this case inaccuracy and discrimination, with the support of independent experts.
- **Testing.** Providers should seek to ensure that foundation models achieve appropriate levels of performance, predictability, interpretability, safety and cybersecurity. Since the foundation model functions as a building block for many downstream AI systems, it should meet certain minimum standards.
- **Documentation.** Providers of foundation models would be required to provide substantial documentation and intelligible usage instructions. This is essential not only to help downstream AI system providers better understand what exactly they are refining or fine-tuning, but also to enable them to comply with any regulatory requirements.

Room for improvement

These new obligations, if adopted in the final AI Act, would be positive steps, but lack detail and clarity, and would consequently rely heavily on harmonised standards, benchmarking and guidelines from the European Commission. They also risk being excessively burdensome. A number of further modifications could be put in place.

Risk-based approach

Applying all obligations to the full extent to every foundation model provider, both large and small, is unnecessary. It might impede innovation and would consolidate the market dominance of firms that already have a considerable lead in FMs, including OpenAI, Anthropic and Google Deepmind⁷. Even without additional regulatory burdens, it might be very hard for any companies outside of this group to match the resources and catch up with the FM market leaders.

A distinction could therefore be made between systemically important and non-systemically important FMs, with significantly lower burdens for the latter. This would be in line with the approach taken by the EU Digital Services Act (DSA), which notes that *"it is important that the due diligence obligations are adapted to the type, size and nature of the ... service concerned."* The DSA imposes much more stringent obligations on certain service providers than on others, notably by singling out very large online platforms (VLOPs) and very large online search engines (VLOEs).

There are two reasons for differentiating between systemic and non-systemic foundation models and only imposing the full weight of mandatory obligations on the former. First, the firms developing systemic foundation models (SFMs) will tend to be larger, and better able to afford the cost of intense regulatory compliance. Second, the damage caused by any deviation by a small firm with a small number of customers will tend to be far less than that potentially caused by an SFM.

There are useful hints in the literature (Bommasani et al, 2023; Zenner, 2023) as to criteria that might be used to identify SFMs, such as the data sources used, or the computing resources required to initially train the model. These will be known in advance, as will the amount of money invested in the FM. These pre-market parameters presumably correlate somewhat with the future systemic importance of a particular FM and will likely also correlate with the ability of the provider to invest in

regulatory compliance. The degree to which an FM provider employs techniques that facilitate third-party access to their foundation models and thus independent verification, such as the use of open APIs, or open source, or (especially for firms that do not publish their source code) review of the code by independent, vetted experts, might also be taken into account. Other, post-deployment parameters, including the number of downloads, or use in downstream services or revenues, can only be identified after the product has established itself in the market.

Lesser burdens

Notwithstanding the arguments for a risk-based approach, even small firms might produce FMs that work their way into applications and products that reflect high-risk uses of AI. The principles of risk identification, testing and documentation should therefore apply to all FM providers, including non-systemic foundation models, but the rigour of testing and verification should be different.

Guidance, perhaps from the European Commission, could identify what these reduced testing and verification procedures should be for firms that develop non-systemic foundation models. Obligations for testing, analysis, review and independent verification could be much less burdensome and intensive (but not less than reasonably stringent) for providers of non-systemic FMs.

This kind of differentiation would allow for a more gradual and dynamic regulatory approach to foundation models. The list of SFMs could be adjusted as the market develops. The Commission could also remove models from the list if they no longer qualify as SFMs.

Use of data subject to copyright

Even though the 2019 EU Copyright Directive provides an exception from copyright for text and data mining (Article 4(1) of Directive 2019/790), which would appear in principle to permit the use of copyrighted material for training of FMs, this provision does not appear in practice to have resolved the issue. The AI Act should amend the Copyright Directive to clarify the permitted uses of copyrighted content for training FMs, and the conditions under which royalties must be paid.

Third-party oversight

The question of third-party oversight is tricky for the regulation of FMs. Is an internal quality management system sufficient? Or do increasingly capable foundation models pose such a great systemic risk that pre-market auditing and post-deployment evaluations by external experts are necessary (with protection for trade secrets)?

Given the scarcity of experts, it will be important to leverage the work of researchers and civil society to identify risks and ensure conformity. A mandatory SFM incident reporting procedure that could draw on an AI incident reporting framework under development at the Organisation for Economic Co-operation and Development⁸ might be a good alternative.

Internationally agreed frameworks

Internationally agreed frameworks, technical standards and benchmarks will be needed to identify SFMs. They could also help document their environmental impacts.

Until now, the development of large-scale FMs has demanded enormous amounts of electricity and has the potential to create a large carbon footprint (depending on how the energy is sourced). Common indicators would allow for comparability, helping improve energy efficiency throughout the lifecycle of an SFM.

Safety and security

Providers of SFMs should be obliged to invest heavily in safety and security. Cyberattacks on cutting-edge AI research laboratories pose a major risk; nonetheless, and despite rapidly growing investments in SFMs, the funding for research in AI guardrails and AI alignment is still rather low. The internal safety of SFMs is crucial to prevent harmful outputs. External security is essential, but it alone will not be sufficient – the possibility of bribes in return for access to models should be reduced as much as possible.

Conclusion

The EU is likely to be a major deployer of generative AI. This market power may help ensure that the technology evolves in ways that accord with EU values.

The AI Act is potentially ground-breaking but more precision is needed to manage the risks of FMs while not impeding innovation by smaller competitors, especially those in the EU. Unless these issues are taken into account in the finalisation of the AI Act, there is a risk of significantly handicapping the EU's own AI developers while failing to install the adequate safeguards.

References

Bender, E., T. Gebru, A. McMillan-Major and S. Shmitchell (2021) 'On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?' *FAccT '21: Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, available at <https://dl.acm.org/doi/10.1145/3442188.3445922>

Bommasani, R., D.A. Hudson, E. Adeli. R. Altman, S. Arora, S. von Arx ... P. Liang (2021) 'On the Opportunities and Risks of Foundation Models', mimeo, available at <https://arxiv.org/abs/2108.07258>

Bommasani, R., K. Klyman, D. Zhang and P. Liang (2023) 'Do Foundation Model Providers Comply with the Draft EU AI Act?' Stanford Center for Research on Foundation Models, available at <https://crfm.stanford.edu/2023/06/15/eu-ai-act.html>

Carugati, C. (2023) 'Competition in generative artificial intelligence foundation models', *Working Paper 14/2023*, Bruegel, available at <https://www.bruegel.org/working-paper/competition-generative-artificial-intelligence-foundation-models>

European Commission (2021) 'Proposal for a regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts', COM(2021) 206 final

OECD (2023) 'AI language models: Technological, socio-economic and policy considerations', *OECD Digital Economy Papers*, Organisation for Economic Co-operation and Development, available at <https://doi.org/10.1787/13d38f92-en>

Zenner, K. (2023) 'A law for foundation models: The EU AI Act can improve regulation for fairer competition', Organisation for Economic Co-operation and Development, forthcoming

The author gratefully acknowledges extensive helpful feedback from Bertin Martens and Kai Zenner.

Endnotes

1 See for example Bender et al, 2021; Bommasani et al, 2021; OECD, 2023.

2 See <https://www.europarl.europa.eu/legislative-train/theme-a-europe-fit-for...>

3 On competition issues raised by foundation models, see Carugati (2023).

4 See <https://oecd.ai/en/network-of-experts/working-group/10836>.

5 See for example Bender et al, 2021; Bommasani et al, 2021; OECD, 2023.

6 See <https://www.europarl.europa.eu/legislative-train/theme-a-europe-fit-for...>

7 On competition issues raised by foundation models, see Carugati (2023).

8 See <https://oecd.ai/en/network-of-experts/working-group/10836>.

Bruegel

Rue de la Charité

33,

B-1210 Brussels

(+32) 2 227 4210

info@bruegel.org

Bruegel.org